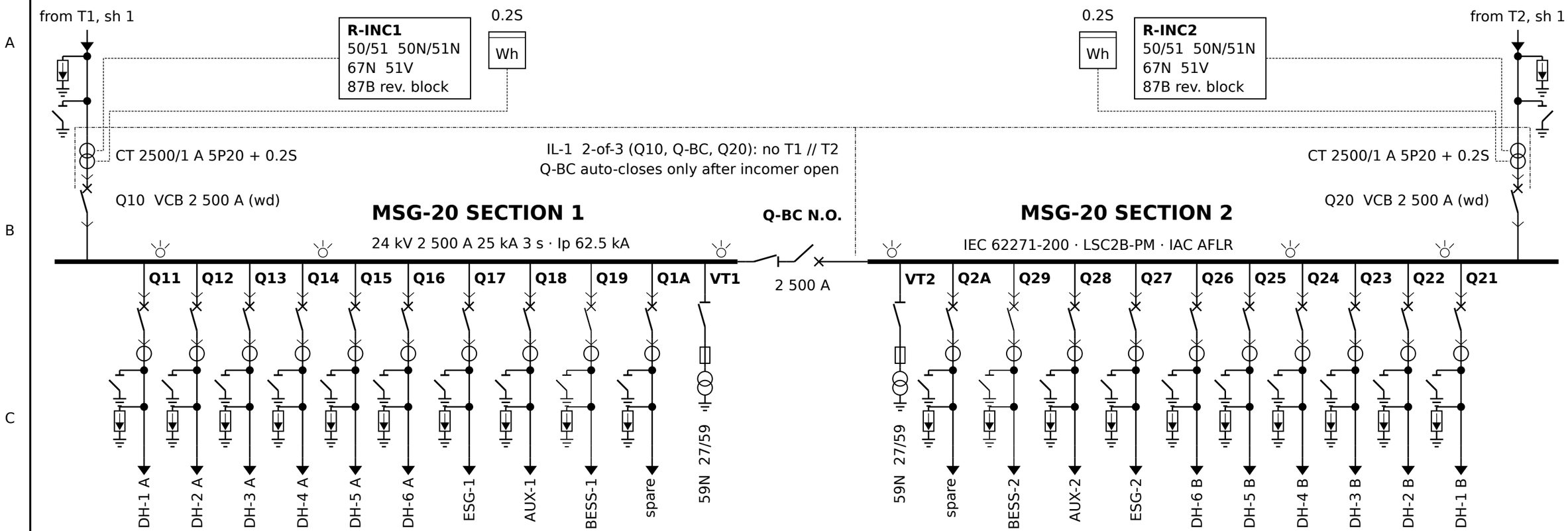
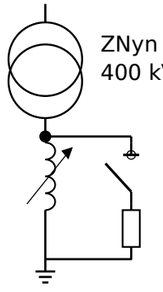


FEEDER SCHEDULE (relay settings per CALC-003)

Bay	Feeds	Device	CT	Protection / metering
Q10 / Q20	T1 / T2 incomer	VCB 2 500 A wd	2500/1 5P20 + 0.2S	51 SI 2 100 A TMS 0.16 50 6 000 A 0.65 s
Q11–16 / Q21–26	DH-1…6 feed A / B	VCB 630 A wd	400/1 5P20 15 VA	51 VI 320 A TMS 0.84 50 2 400 A 0.35 s
Q17 / Q27	ESG: ZNyn + coil	VCB 630 A wd	200/1 5P20	50/51, 51N, 59N
Q18 / Q28	Aux 500 kVA	VCB 630 A wd	100/1 5P20	50/51, 50N/51N
Q19 / Q29	BESS-1 / -2 5.5 MVA	CB 630 A wd	200/1 5P20 + 0.2S	see SLD-004
Q1A / Q2A	spare, equipped	VCB 630 A wd	400/1 5P20	as feeders
Q-BC	coupler (N.O.)	VCB 2 500 A wd	2500/1 5P20	50/51, IL-1

DETAIL B earthing bay Q17 / Q27



ZNyn earthing transformer
400 kVA aux winding (A)

Petersen coil ≥ 50 A, adjustable
10 … 60 A, auto-tuned
‖ 10 A resistor, switched
Ic 22 A/section, 44 A coupled,
+1 A BESS → 67N admittance

NOTES

1 Resonant (compensated) earthing kept; 500 A low-resistance earthing rejected (CALC-003).
2 All feeders: 67N admittance earth-fault relays.
3 Reverse-blocking busbar protection + optical arc protection (50ARC, light + current) trips the section incomer and Q-BC (TODO B18).
4 Hall-board coupler closes only with one hall feeder open (IL-2, SLD-002 sh 1).
5 Every bay: SA 24 kV at the cable end, ES, EPMS 0.5S.
6 DH feeders A → HV-DH1 S1, B → S2 (SLD-002).
7 ESG-1/2: detail B. AUX: 500 kVA station aux.
8 BESS bays Q19 / Q29 → SLD-004 (TODO B22).

P2	2026-10-25	CALC-002/3/4 ratings, IEC symbols, ≥2.5 mm text
P1	2026-10-24	First concept issue

Nordic Hyperscale DC Campus (fictional) · 30 MW IT · SE3
Portfolio D. Wu · public standards and [ASSUMED] data only

20 kV main switchgear MSG-20 — sheet 2
2 sections, coupler, earthing bays, BESS bays
CONCEPT · NOT FOR CONSTRUCTION · scale NTS · A3

Drawing no.	Sheet	Rev
DWU-DC-E-SLD-001	**2 / 2**	**P2**